

Machine Learning for Natural Language Understanding

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The background image shows a top-down view of a workspace. A white notebook is open, showing a pencil sketch of a building or architectural structure. A yellow ruler is placed across the notebook. A black pen and a yellow pencil are visible in the upper left. A small white tag with the word 'LEO' is attached to a string in the upper right. A hand is visible in the lower left, holding a pen. The entire image has a semi-transparent yellow overlay in the center.

Learning

Chapter 02.1

Sources Lecture and Exercise

- Optimization (Derivatives)
 - <https://youtu.be/E5hU0AesA-I>
- Gradient Descent
 - <https://youtu.be/B3w5Zzuqi-E>

Flipped Classroom Questions 1/2

- In which direction does the derivative point?

Optimization (Derivatives)

<https://youtu.be/E5hU0AesA-I>

Classification by Linear Regression

- Linear Regression
$$f(\mathbf{x}_i, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j x_{i,j}$$
$$= \mathbf{x}_i^T \mathbf{w}$$
- Define $y_i = 1$ if $\mathbf{x}_i$ is assigned to class 1 and 0 otherwise
- All new x are assigned to class 1 if $f(x) > 0.5$

Mean Squared Error = Quadratic Loss = L2 Loss

- Supervised Learning:
 - Vector of inputs: $\mathbf{X} = (X_1, X_2, \dots, X_n)$
 - Single output: Y
- Define a function $f(\mathbf{x}, \mathbf{w})$
Mapping $\mathbf{x}$ to y according to parameter vector $\mathbf{w}$
- Define a loss function $\text{loss}(y, f(\mathbf{x}, \mathbf{w})) = (y - f(\mathbf{x}, \mathbf{w}))^2$
Typically quadratic loss

Squared Error

- Given numeric target values and numeric outputs, we can define a loss function ℓ , which measures **how severe** a certain error is.
- Popular (standard) choice is the **squared error loss**:

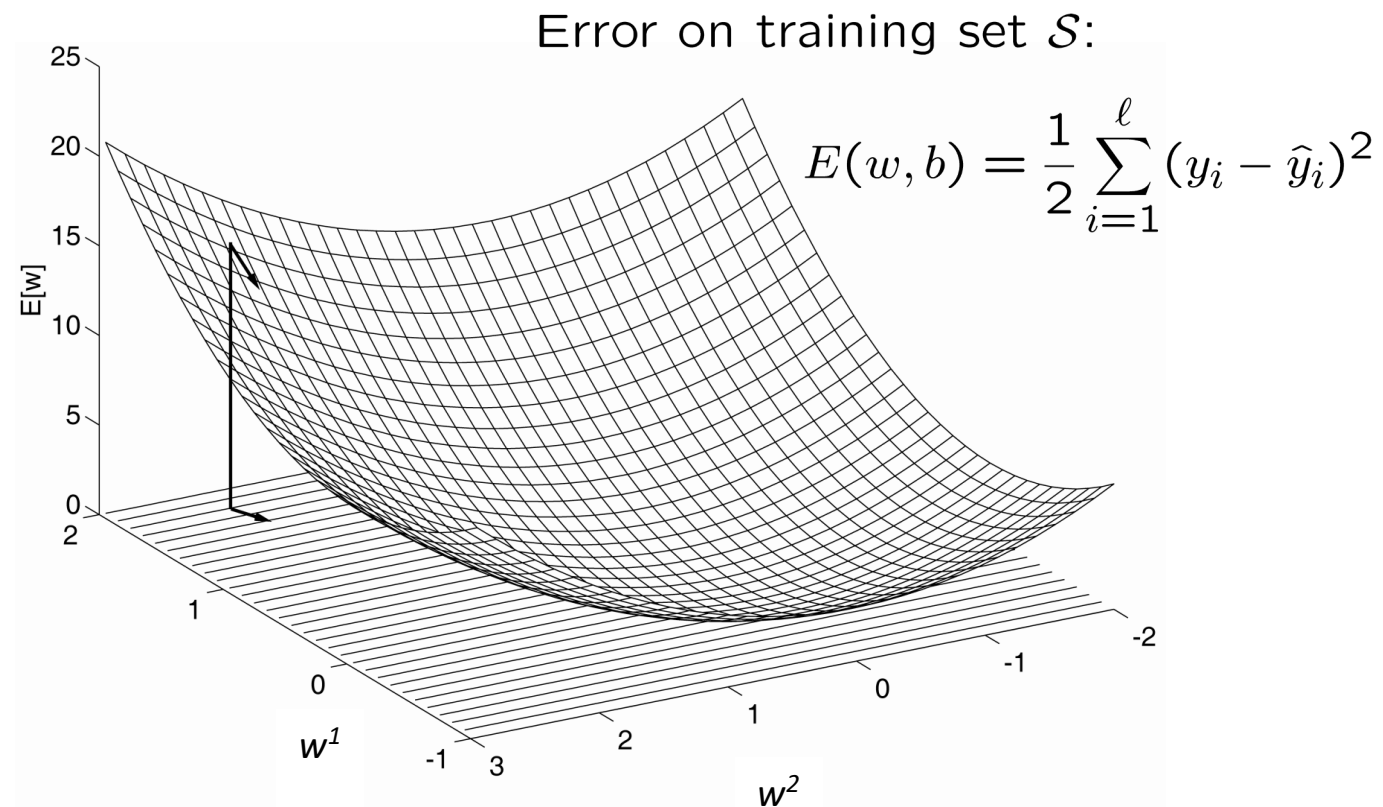
Error on individual example:

$$\ell(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$$

Error on training set $\mathcal{S}$:

$$E(w, b) = \frac{1}{2} \sum_{i=1}^{\ell} \underbrace{(y_i - \hat{y}_i)^2}_{\ell(y_i, \hat{y}_i)} = \frac{1}{2} \sum_{i=1}^{\ell} (y_i - (\langle w, x_i \rangle + b))^2$$

Error Surface and Gradient Descent



Direction of steepest descent along the error surface

Figure taken from Mitchell (1997)

Gradient Descent

- **Gradient** – direction in weight space that produces the steepest increase in error (partial derivatives of error wrt to the individual components of current weight vector):

$$\nabla E(w) = \left[\frac{\partial E}{\partial w^1}, \dots, \frac{\partial E}{\partial w^n} \right]$$

$$\nabla E(b) = \left[\frac{\partial E}{\partial b} \right]$$

- The update step then becomes (α = learning rate):

$$w \leftarrow w - \alpha \nabla E(w)$$

$$b \leftarrow b - \alpha \nabla E(b)$$

Derivation of Gradient for Squared Error

$$\begin{aligned}\frac{\partial E}{\partial w^j} &= \frac{\partial}{\partial w^j} \frac{1}{2} \sum_{i=1}^{\ell} (y_i - \hat{y}_i)^2 \\ &= \frac{1}{2} \sum_{i=1}^{\ell} \frac{\partial}{\partial w^j} (y_i - \hat{y}_i)^2 \\ &= \frac{1}{2} \sum_{i=1}^{\ell} 2(y_i - \hat{y}_i) \frac{\partial}{\partial w^j} (y_i - \langle w x_i \rangle) \\ &= - \sum_{i=1}^{\ell} (y_i - \hat{y}_i) x_i^j \\ \nabla E(w) &= - \sum_{i=1}^{\ell} (y_i - \hat{y}_i) x_i\end{aligned}$$

NB: Derivation is analogous for $b = w^0$ (by assuming $x^0=1$ for any input)

Gradient Descent

<https://youtu.be/B3w5Zzuqi-E>

Gradient Descent Training for Perceptrons (see next lecture)

Algorithm 1 Gradient Descent (Batch)

Input: data $\mathcal{S}$; learning rate α

Initialize: $w \leftarrow 0, b \leftarrow 0$

1: **repeat**

2: $\Delta w \leftarrow 0, \Delta b \leftarrow 0$

3: **for** $i = 1, \dots, \ell$ **do**

4: compute $\hat{y}_i = \langle w, x_i \rangle + b$

5: $\Delta w \leftarrow \Delta w + \alpha (y_i - \hat{y}_i) x_i$

6: $\Delta b \leftarrow \Delta b + \alpha (y_i - \hat{y}_i)$

7: **end for**

8: $w \leftarrow w + \Delta w$

9: $b \leftarrow b + \Delta b$

10: **until** some termination criterion

11: **return** w, b

← imagine this to be a dummy feature that equals 1 for all inputs.

← w and b are updated after all examples have been processed

← see later



Evaluation

Chapter 02.2

Sources Lecture and Exercise

- Hold-out data
 - <https://youtu.be/kFaoSM-wn4k>
- Overfitting / Underfitting
 - <https://youtu.be/x-OxfOHPLsc>
- Workflow (Train vs Validation vs Test)
 - https://youtu.be/xAo77BqF_nI
- (Deep Dive: Model evaluation
 - <https://arxiv.org/pdf/1811.12808.pdf>
- Evaluation (simple metrics for binary classification)
 - <https://youtu.be/CTQm8T6PZKg>
- BLEU
 - <https://youtu.be/M05L1DhFqcw?si=ZOjNn62HYADD6Jh2>
- Perplexity
 - https://youtu.be/NURcDHhYe98?si=T4jzZ9_Sk2Z4TC54

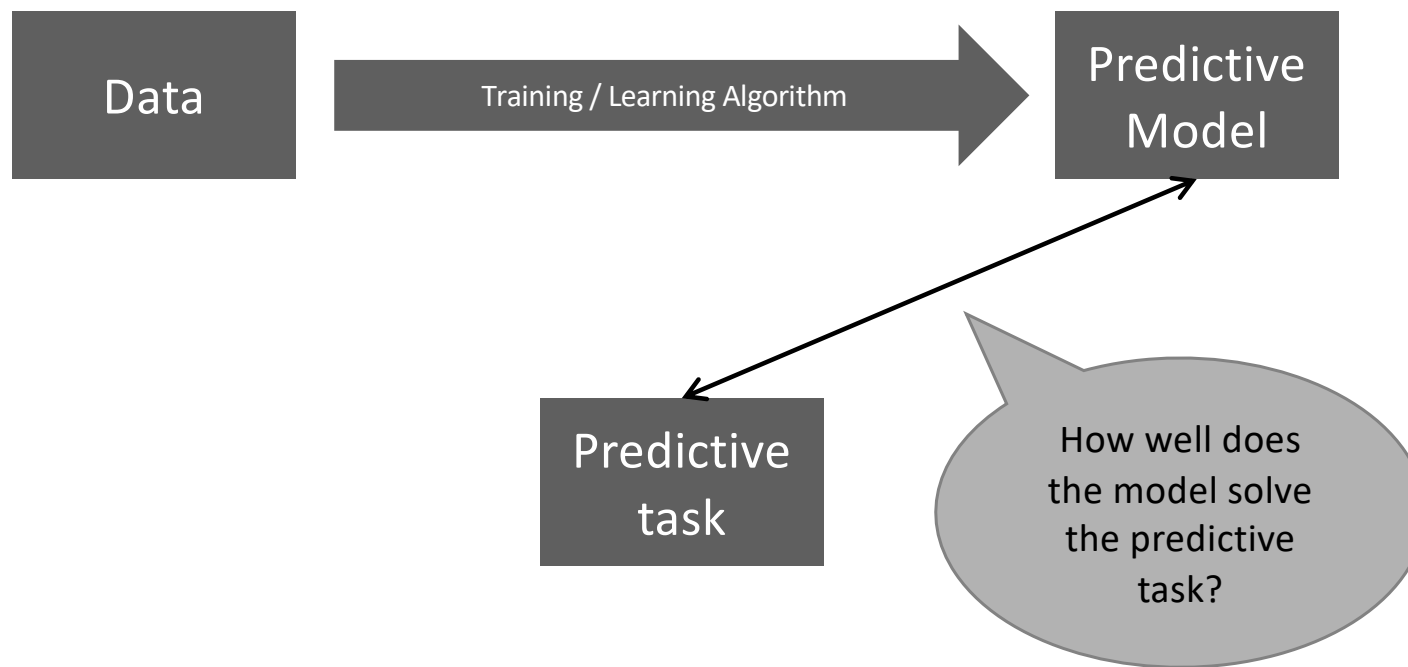
Flipped Classroom Questions 2/2

- Why should I split my labeled data?
- What is overfitting? Name one potential reason for overfitting.
- How is the Tuning Set also called?
- What are elements of a confusion matrix?
- What does a probabilistic classifier produce?
- What are limits of binary classification metrics when applied in NLP?

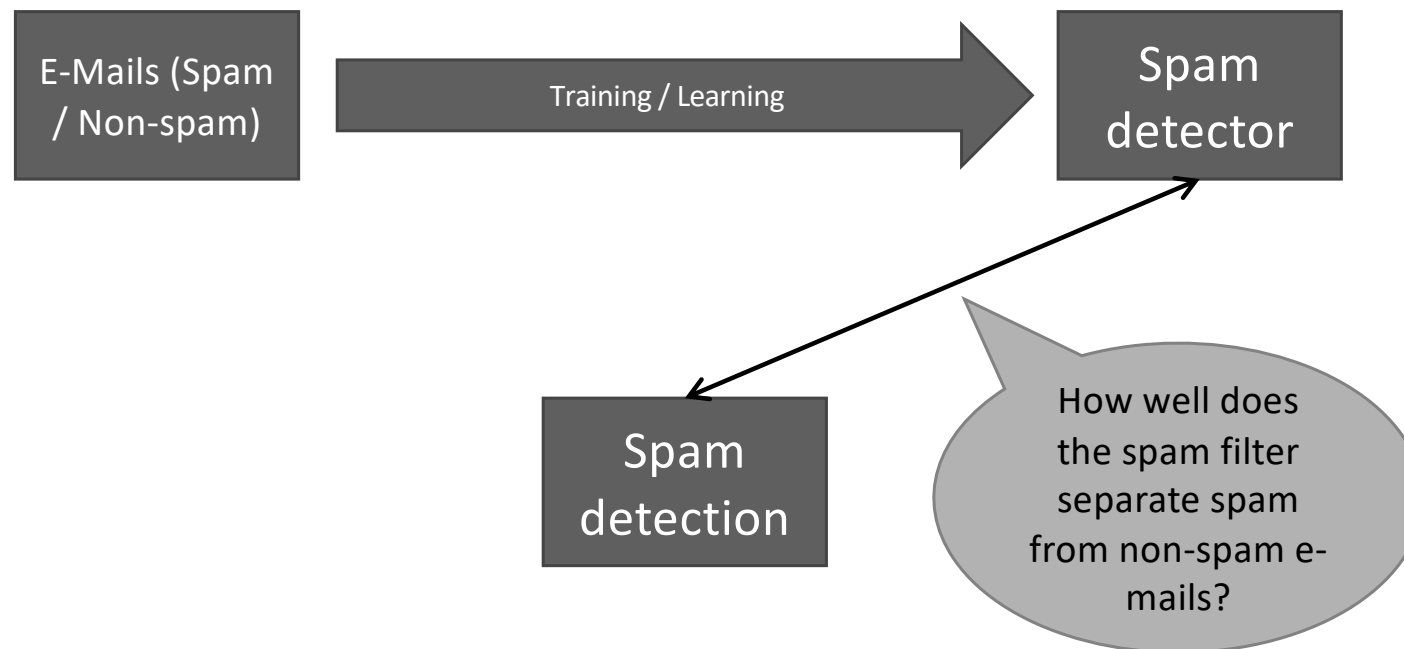
Hold-out data

<https://youtu.be/kFaoSM-wn4k>

Predictive Models - The big picture



Example – Spam detection



Validation Goal

How well does the model
perform on new (= unknown)
data?

a.k.a.: “Generalization”

Choice of Test Data: Attempt I

- Possibility 1: Resubstitution Validation
 - Test Set = Training Set
 - Use all the available labeled data for training
 - Evaluate method by classifying the training set using
- Problems:
 - Model which performs best is model which best represents the training data
 - Performance on new data is not known

Training Data = Test Data

Choice of Test Data: Attempt II

- Possibility 2: Training-/Testsplit
 - Split available data into two subsets: Training and Test Set (Hold-out set)
 - Train on training data, evaluate on test data
 - Typical split: 66% training data, 33% test data (for „small“ data sets)
- Advantages:
 - Evaluation on data which is not used for training
- Problems:
 - Not all the data is used for training
 - Evaluation results strongly depend on choice of training and test set

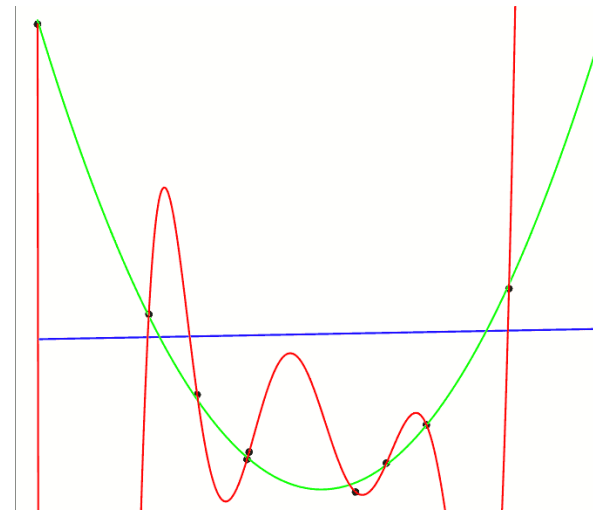


Overfitting / Underfitting

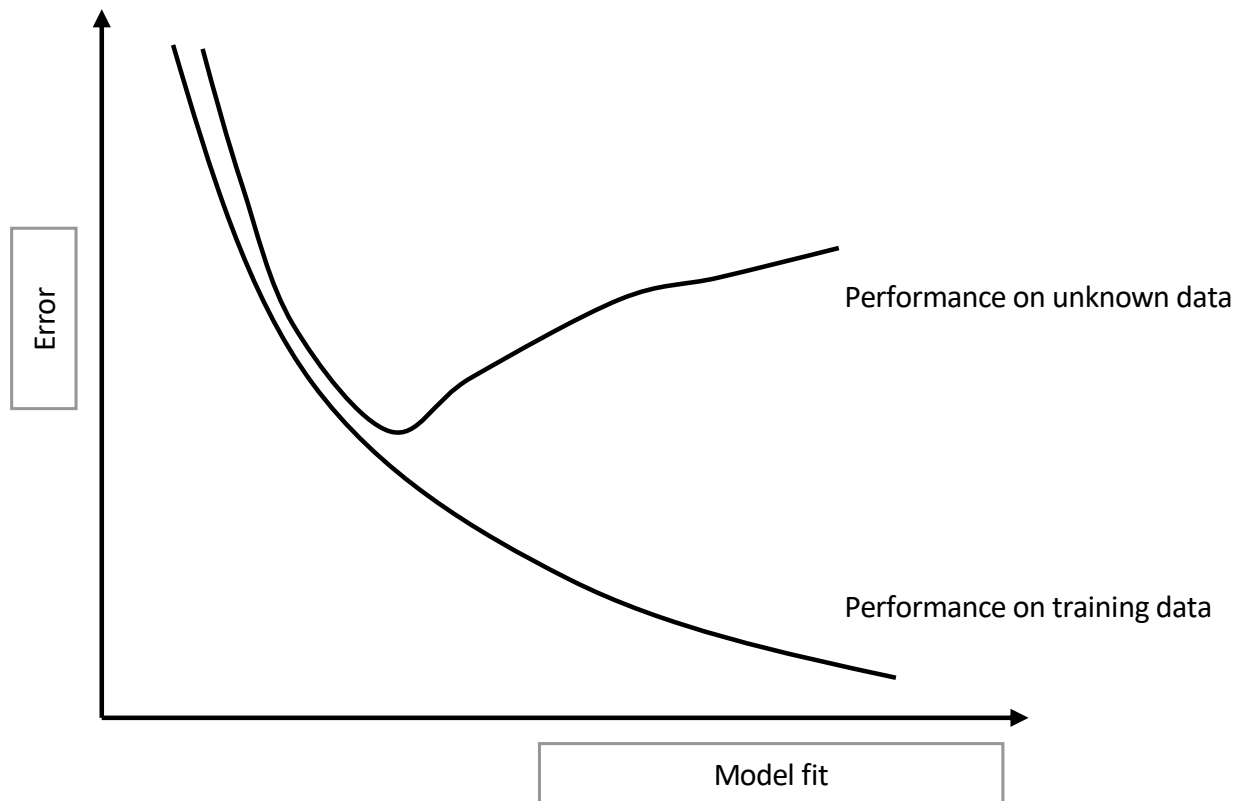
<https://youtu.be/x-OxfOHPLsc>

Overfitting

- Model is constructed based on sample of data (black dots) from a true function (green)
- Many prediction models may be derived from the data
- Models which are too simple are underfitting the data (blue line), i.e. a more complex model will describe the data better
- Too complex models overfit the data (red line), they memorize the training data, i.e. they perform well on the training data, but poorly on unknown data



Overfitting

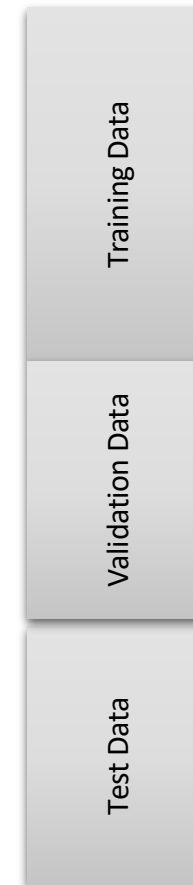


Workflow (Train vs Validation vs Test)

https://youtu.be/xAo77BqF_nl

Validation Set

- Validation-/tuning or development set used for tuning parameters or selecting between different models
- not directly used for training
- But not used for final testing!



Evaluation (simple metrics for binary classification)

<https://youtu.be/CTQm8T6PZKg>

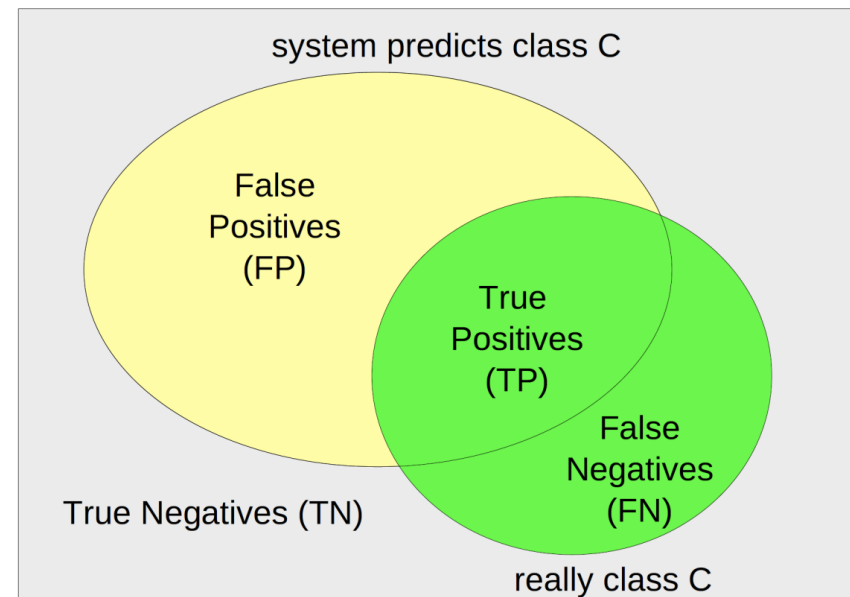
Evaluation measures

- Requirements:
 - Describe quality of the learned model in terms of performance on evaluation data
 - Show weaknesses of the model with respect to intended application
- Example: Spam classification
 - Measures should show high quality for a classifier who detects most spam, but does not classify any normal e-mail as spam

Measures for Classification: Confusion matrix

- Confusion matrix for 2 classes

		Predict C?	
		Yes	No
Really C?	Yes	TP	FN
	No	FP	TN



- May be generalized to multiclass case
- Good model:
 - High values on diagonal
 - Low FP and FN

Example: Spam classification

- Collection of e-mails
 - 1000 non-spam e-mails
 - 1000 spam e-mails
- Classifier for automatic spam detection:
 - Of the 1000 non-spam e-mails
 - 950 classified as non-spam
 - 50 classified as spam
 - Of the 1000 spam e-mails
 - 100 classified as non-spam
 - 900 classified as spam

Measures for Classification: Error and Accuracy

- General criteria

- Classification Error

$$\frac{\text{errors}}{\text{total}} = \frac{FP + FN}{TP + FN + FP + TN}$$

- Accuracy (= 1 - Error)

$$\frac{\text{correct}}{\text{total}} = \frac{TP + TN}{TP + FN + FP + TN}$$

- Problem: Meaningless for unbalanced datasets

- Prediction of occurrence of earthquake

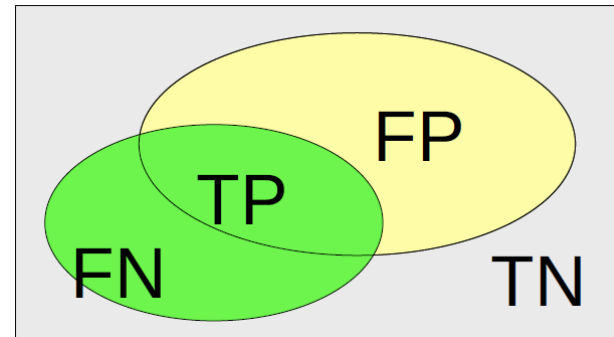
- Rare event
- Classifier with high accuracy: always predict no
- → FN should have higher cost than FP

- Internet search: Is page relevant for user?

- 99,99999% of pages are irrelevant for a given query
 - best answer is always no
- → Use metrics which do not use TN (e.g. recall / precision)

Measures for Classification:

- False Alarm Rate (FA)
= False Positive Rate (FPR)
= $FP / (FP + TN)$
 - % of the negative instances which are classified as positive
- Miss Rate (MR) = False Negative Rate = $FN / (FN + TP)$
 - % of the positive instances which are classified as negative
- Recall = True Positive Rate (TPR) = Sensitivity = $TP / (TP + FN)$
 - % of positive instances which are correctly classified (1 - Miss Rate)
- Precision = $TP / (TP + FP)$:
 - % of instances classified as positive which are really positive
- Specificity = True Negative Rate = $TN / (TN + FP)$
- F-measure (harmonic mean of precision and recall)
= $2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$



Example: Document search

The screenshot shows a Google search interface with the query 'cross validation'. The search bar includes the Google logo, the text 'cross validation', a search button, and a link to 'Erweiterte Suche'. Below the search bar, it indicates 'Suche' and 'Ungefähr 5.150.000 Ergebnisse (0,09 Sekunden)'. On the left, there are filters for 'Alles', 'Bilder', 'Maps', 'Videos', 'News', 'Shopping', 'Bücher', and 'Mehr'. Below these are filters for 'Karlsruhe' (Standort ändern), 'Das Web' (Seiten auf Deutsch, Seiten aus Deutschland, Übersetzte Seiten), 'Alle' (Letzte Stunde, Letzte 24 Std., Letzte Woche, Letzter Monat, Letzte 2 Monate, Letztes Jahr, Zeitraum festlegen...), and 'Alle Ergebnisse' (Verwandte Suchanfragen, Mehr Optionen). The main results area shows several links, including Wikipedia entries for 'Cross-validation', 'Cross validation (statistics)', and 'Kreuzvalidierungsverfahren', as well as a PDF from Arizona State University titled 'Cross-Validation - public.asu.edu'.

Recall: How many of the relevant documents are returned as relevant?

Precision: How many of the returned documents are really relevant?

Towards advanced NLP metrics

<https://youtu.be/M05L1DhFqcw?si=ZOjNn62HYADD6Jh2>

https://youtu.be/NURcDHHYe98?si=T4jzZ9_Sk2Z4TC54

Evaluating Generative LMs is tough (see later lectures)

- Next word prediction
 - Extend binary classification metrics to multiclass metrics
 - Cross entropy loss, Perplexity
- Comparing a generated text to a gold standard text
 - BLEU, ROUGE, METEOR,...
- Asking a human expert to score the generated text
 - Likert scale, Pairwise ranking,...

“The future of language is about the development of language models...”¹

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